

Bing Fan

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RESEARCH INTERESTS

- **Video understanding:** egocentric visual query localization and segmentation, ego–exo cross-view object correspondence, spatio-temporal omni-object video grounding, and tissue point tracking in endoscopic surgical video.
- **Multimedia forensics:** detection, localization, and attribution of AI-generated and manipulated media; DeepFake anti-forensics and adversarial perturbation defense.

EDUCATION

Ph.D. in Computer Science and Engineering, University of North Texas, USA (Advisor: Prof. Heng Fan)	2024 – 2027
M.S. in Cyberspace Security, Nanchang University, China (Advisor: Prof. Feng Ding)	2021 – 2024
B.S. in Software Engineering, Nanchang University, China	2017 – 2021

HONORS & AWARDS

1st Place in both 2D and 3D Tracking Accuracy, MICCAI 2026 STIR Challenge (Team SurgTrackV2)	2026
Chinese National Scholarship	2023

PUBLICATIONS

First-Author / Co-First-Author

* co-first authors

- [1] **B. Fan**, M. Li, H. Zhang, S. Dong, N. P. Mareedu, W. Shi, Y. Feng, Y. Huang, H. Fan, “Towards Visual Query Segmentation in the Wild.” *Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] **B. Fan**, Y. Feng, Y. Huang, H. Fan, “Exploiting Textual Semantics for Robust Cross-View Object Correspondence.” *Neural Information Processing Systems (NeurIPS)*, 2026.
- [3] **B. Fan**, Y. Feng, Y. Tian, J. C. Liang, Y. Lin, Y. Huang, H. Fan, “PRVQL: Progressive Knowledge-Guided Refinement for Robust Egocentric Visual Query Localization.” *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [4] Y. Hu*, **B. Fan***, X. Gu, H. Ren, D. Liu, H. Fan, L. Zhang, “Robust Ego-Exo Correspondence with Long-Term Memory.” *Neural Information Processing Systems (NeurIPS)*, 2025.
- [5] **B. Fan**, F. Ding, G. Zhu, J. Huang, S. Kwong, P. Atrey, S. Lyu, “Generating Higher-Quality Anti-Forensics DeepFakes with Adversarial Sharpening Mask.” *ACM Transactions on Multimedia Computing, Communications and Applications (TOMM)*, vol. 21, 2025.
- [6] **B. Fan**, S. Hu, F. Ding, “Synthesizing Black-Box Anti-Forensics DeepFakes with High Visual Quality.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

Collaborative Work

- [7] J. Yao, X. Deng, X. Gu, M. Dai, **B. Fan**, Z. Zhang, Y. Huang, H. Fan, L. Zhang, “OmniSTVG: Toward Spatio-Temporal Omni-Object Video Grounding.” *International Conference on Learning Representations (ICLR)*, 2026.
- [8] Y. Zhou, X. He, K. Lin, **B. Fan**, F. Ding, B. Li, “Breaking Latent Prior Bias in Detectors for Generalizable AIGC Image Detection.” *Neural Information Processing Systems (NeurIPS)*, 2025.
- [9] X. He, Y. Zhou, **B. Fan**, B. Li, G. Zhu, F. Ding, “VLForgery Face Triad: Detection, Localization and Attribution via Multimodal Large Language Models.” *Neural Information Processing Systems (NeurIPS)*, 2025.
- [10] F. Ding, **B. Fan**, Z. Shen, K. Yu, G. Srivastava, K. Dev, S. Wan, “Securing Facial Bioinformation by Eliminating Adversarial Perturbations.” *IEEE Transactions on Industrial Informatics (TII)*, 19(5): 6682–6691, 2022.